Koa Shen

Mechanical Engineer
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Dynamics, Statics, Mechanics of Materials, Calculus-based Physics, MATLAB. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Built an open-source Python simulation framework on Drake for kinematics and collision detection in tightly enclosed X-ray beamline assemblies, avoiding commercial software that costs about $25k per seat per year.
- Modeled a 30-DOF, 305-body assembly of linear, rotary, and tip-tilt stages; collision checks run in milliseconds and full models load in about 30 seconds.
- Built a CAD-to-simulation pipeline using OpenCascade for STEP import, CoACD convex decomposition for collision geometry, and MeshCat for 3D visualization.
- Verified collision results with a secondary mesh triangle-triangle distance query, and compared models with the physical assembly to catch outdated CAD before it skewed results.
- Identified at least five real pinch points and 10+ actionable design changes; built a 15-stage reusable catalog now being adopted on three LCLS assemblies, with path planning and EPICS integration in development.
- Applied simulation findings to an optics assembly redesign and checked stage and mount load cases with statics and dynamics calculations.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Used hand calculations and FEA to compare CFRP, GFRP, and aluminum steering-wheel baseplates; FEA-guided iterations cut mass from 5 lb to 1.7 lb; tested for strength and rules compliance.
- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the car passed FSAE technical inspection and placed 2nd among UC teams at 2026 Michigan FSAE Electric.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding bearing interfaces within 0.0005 in.

PROJECTS

Closed-Loop Cycloidal Actuator | C++, Personal Project — Summer 2026 - Present

- Wrote C++ firmware in PlatformIO for TMC2209 stepper control, including phase-based bring-up, step/dir pulse generation, and real-time CSV telemetry of time, step count, and measured angle.
- Implemented closed-loop correction from a 12-bit AS5600 encoder; bench-validated automatic position-error compensation and missed-step detection by comparing commanded and measured motion.
- Designed a 15:1 cycloidal reducer packaged within the motor's 42 mm face as the first joint of a planned 6-DOF arm; Drake kinematic simulation and trajectory optimization are next.

Interactive Exhibit Overhauls | Exploratorium, Maintenance Technician — Summer 2025

- Diagnosed failures across 100+ public exhibits and led three major overhauls, including a load-spreading bezel redesign that avoided $3,000+ in replacement costs.

TECHNICAL SKILLS

Simulation and Geometry: Drake, rigid-body kinematics, collision detection, convex decomposition, mesh distance queries, OpenCascade, MeshCat, 3D geometry
Programming: C++, C, Python, MATLAB, PlatformIO, Arduino, Git/GitHub
Engineering: Dynamics, statics, FEA, Solid Edge, SolidWorks, Teamcenter PDM, GD&T, CNC machining